THE PALEO MANIFESTO

ANCIENT WISDOM FOR LIFELONG HEALTH



JOHN DURANT

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THE PALEO MANIFESTO

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THE PALEO MANIFESTO

Ancient Wisdom for Lifelong Health

JOHN DURANT

with contributions by

Michael Malice



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First Edition

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3

RISE AND FALL

PALEOLITHIC AGE

◂ *2.6 million years ago to 10,000 years ago* ▸

Five years after graduating, I found myself back on the Harvard campus in a familiar situation: late to meet with a professor. My destination? Harvard's Peabody Museum of Archaeology and Ethnology.

The Peabody boasts a collection of over six million ancient artifacts from around the world: a shark tooth spearhead from the South Pacific; Mayan hieroglyphics engraved on giant limestone slabs; a Native American whistle carved out of eagle bone collected by Lewis and Clark on their legendary transcontinental expedition.

The Peabody is also where Harvard stores its osteological collections: bones. *Lots* of bones. Neanderthals from Europe, mummified remains from South America, chimpanzees from Africa. The collections contain fossils famous for documenting the emergence of human beings, as well as skulls with uncommon deformities used to teach students about skeletal development. Delicate and rare, the bones in the collections are locked away in an archive, not on display to the public.

I was meeting with Dr. Daniel Lieberman, chair of the Department of Human Evolutionary Biology. Bearded and bespectacled, he looked just like a professor should; he was the ur-professor. Dr. Lieberman studies bones, both dead bones (paleoanthropology) and living bones (biomechanics). He earned tenure by studying the

human head, but he earns mention in ESPN stories for his research on barefoot running.

The entire field of human evolutionary biology, Lieberman explained, is dedicated to answering a simple question: *Why are humans the way we are?*

Why are our teeth shaped the way they are—and what does that say about the human diet? Why is the foot arched—and what does that say about the biomechanics of walking and running? Why do humans sweat so much more than other mammals? Why do humans have such large and powerful brains? Embedded in the blueprint of our bodies are clues to the deep history of humanity—and many of those clues point to the Paleolithic Age.

Paleo means "old," *lithic* means "stone," and the Paleolithic begins with the first known stone tools used by hominins about 2.6 million years ago. (The term "hominins" refers to humans and our human-like ancestors such as *Homo erectus, Homo habilis,* and *Australopithecus*.) The stone tools were found in a small area of Eastern Africa in northern Tanzania, near the Serengeti, in a place known as Olduvai Gorge. Paleoanthropologists refer to the area as the "Cradle of Mankind."

The earliest stone tools were rudimentary and didn't change for long stretches of time. They might easily have remained an isolated use of technology by a specific species, the primate equivalent of beaver dams. But by the Upper Paleolithic—roughly 50,000 years ago—something important had changed: artifacts become more diverse and complex, and artistic decoration becomes easy to recognize. *Humanity* becomes easy to recognize. Scholars debate whether the Upper Paleolithic Revolution was the result of rapid changes or more gradual ones, but no one questions that those changes produced a truly revolutionary species: us.

Within a relatively short period, humans expanded into nearly every habitable land and island across the globe, from the barren Arctic tundra to remote atolls of the South Pacific. We displaced all other lineages of hominins (Neanderthals, Denisovans), domesticated

another species for the first time (wolves), and hunted dozens of species to extinction. No species had ever established such widespread dominance over dry land so quickly and so thoroughly.

Humanity's explosion onto the world stage was made possible by the expansion of our most defining trait: the brain.

At the beginning of the Paleolithic, our hominin ancestors had modestly larger brains than chimpanzees. Over the next two and a half million years, brain size more than doubled. As a percentage of body mass, humans now have one of the largest brains of any species. More important than raw size is what we can do with our brains, such as communicate with complex language and learn through culture. Culture gave humans the behavioral flexibility to dramatically change how we lived, even within the span of a generation, and then pass that knowledge on to the next generation. It's what enabled hunter-gatherers to adapt to habitats across the globe with such incredible speed.

Dr. Lieberman and his colleagues are trying to figure out what factors influenced the emergence of human beings, and I got to peek into some of the experiments going on in the department.

Dr. Lieberman and his collaborator, Dr. Dennis Bramble, have proposed that bipedal humans are adapted to long-distance walking and endurance running, which our ancestors may have used for scavenging and persistence hunting (running an animal to exhaustion under the hot savannah sun). Dr. Lieberman showed me his running lab, and I got to try out one of the world's most expensive treadmills. You can't buy treadmills like this—Lieberman and his team customized it themselves. Two rubber treads—split, to record left and right feet independently—covered steel force plates, which measured the location and intensity of forces thousands of times a second. A plastic tubing contraption strapped over a runner's mouth measured oxygen consumption, enabling his team to calculate energy consumption. A series of high-speed infrared cameras tracked body movements from little reflective markers taped to the runner's body, similar to the technology used to track movements for CGI effects in movies.

In the same room I opened a freezer containing the butchered remains of a goat. They were studying the jaw forces and energy required to chew raw meat, a privilege that would be bestowed on a few lucky undergraduates. Paleoanthropologists don't agree on how our ancestors hunted before the advent of stone-tipped spears and projectile weapons, but early Paleolithic sites are often littered with bones that bear the distinct markings of an animal having been butchered. Whether by scavenging, ambush hunting, or another method they succeeded.

Changes in diet played an important role in the expansion of the human brain. A big brain is powerful, but it's also expensive. The human brain is about 2 percent of our body mass but consumes roughly 20 percent of our energy. To feed a power-hungry brain, our ancestors had to secure high-quality sources of calories and nutrients. Food sources naturally tend to fall on a spectrum from "easy to obtain, but low nutritional value" (grass, leaves) to "hard to obtain, but high nutritional value" (mammoth, tubers). Our ancestors' diet shifted away from a largely herbivorous or frugivorous diet, akin to the diets of many of the great apes, and toward an omnivorous diet—first, by adding more roots and tubers, and then by introducing more meat.

Just down the hall, scientists were in a "kitchen" heating different types of food to exact temperatures for precise periods of time. Dr. Richard Wrangham, one of Dr. Lieberman's colleagues, has been studying the role of cooking on the expansion of the brain.

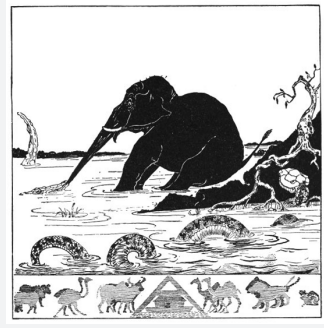
Cooking acts like an external form of digestion, breaking down tough plant and muscle fibers. As a result, cooked food yields more calories than raw food—plus it kills parasites and improves the edibility of a much wider range of wild foods. The earliest evidence of cooking appears roughly a million years ago, though scholars still debate when it became a regular and widespread practice. Once hominins wielded fire, the energy constraints on brain size were relaxed even further.

It's hard to pinpoint the exact benefits of an incrementally larger

JUST-SO STORIES

The Elephant's Child sat there for three days waiting for his nose to shrink. But it never grew any shorter . . . For, O Best Beloved, you will see and understand that the Crocodile had pulled it out into a really truly trunk same as all Elephants have to-day.

—Rudyard Kipling,
Just So Stories (1902)



Humans are good storytellers, but not all good stories are true. The elephant didn't actually get its trunk from a tug-of-war with a crocodile. When discussing evolutionary hypotheses, it's important to keep a few things in mind.

1. **What constitutes an "adaptation" is complicated.** Most adaptations are a trade-off between multiple uses. The female pelvis must allow women to walk as well as give birth.
2. **Evolution selects for reproductive success, not necessarily health or longevity.** NFL players get lots of babes, but they get injured a lot and don't live as long.
3. **There was no single "natural habitat" for humans.** The African savannah was a very important habitat, but modern humans bear the imprint of many places and times—from the beginning of life right up to the present.
4. **Evolution hasn't stopped.** Evolution didn't stop 10,000 years ago. (In fact, it has actually accelerated.)
5. **Evolution is amoral.** Just because something is natural doesn't make it morally good ("the naturalistic fallacy"). Conversely, just because something is morally desirable doesn't make it true ("the moralistic fallacy").

Just as evolution is a continual process, so too is our knowledge of it. Evolutionary theory is great for generating hypotheses, but those hypotheses must be tested against evidence.

Source: *The Story of the Human Body: Evolution, Health, and Disease* by Daniel Lieberman

brain, but various cognitive functions must have given hominins a competitive advantage in foraging, social cooperation, language use, and sexual selection (i.e., impressing the opposite sex). However it happened, it did—and once we became big-brained hunter-gatherers, humans were hard to stop. But the most radical shift in lifestyle took place when some hunter-gatherers gradually settled down and started farming.

The Agricultural Revolution was the result of a revolutionary growth technology called domestication, a gradual form of genetic engineering. Domestication is a multigenerational process where humans breed other species to meet our needs, whether it is as a source of food (wheat), companionship (dogs), or even beauty (roses). Among animals, humans domesticated the wild ancestors of dogs, sheep, pigs, goats, cows, chickens, and horses. Among plants, humans domesticated the wild precursors of wheat, barley, rye, squash, maize, beans, millet, rice, soy, and potatoes.

Hunters of wild prey became *herders* of domesticated animals. *Gatherers* of wild plants became *farmers* of domesticated crops. "Hunter-gatherers" became "herder-farmers." Our diverse, omnivorous diet as hunter-gatherers became heavily grain-based, contributing to an overall decline in health—and I was about to hold the evidence in my own hands.

"Okay, let's go see some skulls," said Dr. Lieberman, as he led me out of the department and into the museum.

We walked up to an old wooden door. Probably part of the original construction in the late nineteenth century, it had been retrofitted with an electronic security system. Dr. Lieberman swiped a key fob past it, the lock clicked open, and we walked into a brightly lit room with laboratory equipment lining one wall. It felt like moving from historic to modern, but we were actually moving from historic to *pre*-historic.

The archive felt like a cross between a library and a morgue. It was lined with shelves, which were filled with boxes of bones. Handwritten

words faced outward: "Natufian—El Wad"; "Chimpanzee—Liberia." The labels served the dual role of title and tombstone.

Dr. Lieberman handed me a pair of latex gloves.

"Here, put these on."

The gloves weren't there simply to protect the ancient remains from me; they were also there to protect me from the ancient remains, since many had been preserved in nasty chemicals.

Dr. Lieberman pulled a box off the shelf, carried it over to a table, and took off the lid. He gently lifted up a skull.

"This is Skhul V. This guy is famous. He was a hunter-gatherer living more than 80,000 years ago in the Levant. That's modern-day Israel. He's one of the earliest anatomically modern *Homo sapiens* ever recovered. Here, you can hold his skull. There's only one rule."

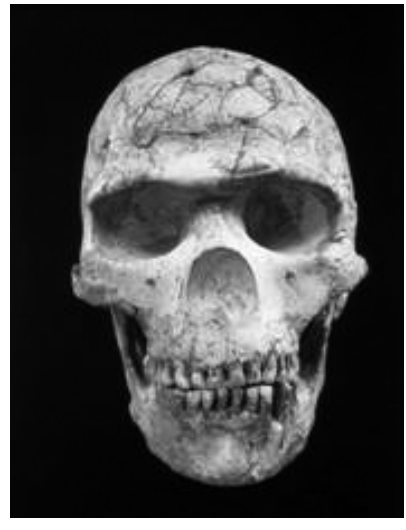
Dr. Lieberman paused and looked directly at me.

"Use two hands."

It was a command, not a suggestion. This is not something you want to drop.

With that he gingerly passed me Skhul V.

Looking at a human skull creates an optical illusion: it looks like the skull is smiling. The brain interprets the visible teeth and upswept jawbone as an upturned mouth. Not only did this phenomenon create the creepy effect that Skhul V was somehow alive, but he seemed downright cocky, brimming with a confidence that even the grave couldn't shake.



And what a grin this guy had. What an *amazing* grin.

"Notice anything?" Dr. Lieberman asked. "Look at the teeth. They're straight. And no cavities. His wisdom teeth came in just fine. Humans, like all animals, have evolved teeth that are well suited to

their natural diet. An infected tooth can easily kill you, and there were no dentists in the Paleolithic."

Nearly one hundred thousand years before dentists and orthodontists, this hunter-gatherer had a strong, straight set of chompers. Skhul V challenged much of what I'd been taught about the history of human health.

"Now, look," Dr. Lieberman continued, "hunter-gatherers didn't have *perfect* teeth. This guy has well-worn teeth, and he's actually missing one due to an abscess. So don't stop going to the dentist. But wait until I show you the skulls of early farmers—a lot of them would need to get fitted for dentures."

"So what's the secret? Eat less sugar?" I asked.

"Well, yes, but healthy teeth depend on a variety of factors," Dr. Lieberman explained. "First, yes, it matters what you eat. Amylase in your saliva breaks down carbohydrate into sugar in your mouth. Bacteria feed on the sugar and produce acid that wears away the enamel on teeth, giving you cavities. We'll see what happened to the early farmers who started eating a starchier diet."

To figure out what humans used to eat, teeth are a good place to start. Not only do teeth fossilize well, but they're the first point of contact between the food we eat and our body, the first part of our internal digestive tract. And if our teeth aren't well adapted to a particular food, it's unlikely the rest of our digestive tract is. But whatever Skhul V was eating, his teeth seemed to be up to the challenge.

"It also matters how tough your food is to chew," Dr. Lieberman continued. "When you put force on bones, they grow bigger and stronger. People back then ate tougher foods, they put larger forces on their jaw, and thus they had jaws large enough to actually fit all of our teeth. And those bite forces may have helped our teeth come in straight."

"And since we all eat such soft foods these days?" I asked.

"Smaller jaws."

It felt oddly insulting to hear him say that. In a sense he was

pointing out that my growth had been stunted in childhood. I'm *deformed*. And not just me, but most modern people.

Dr. Lieberman took the skull and put it back in the box, then pulled out a femur and held it up.

"I don't know if you've seen many femurs, but I have, and this is quite a femur. He almost certainly has much thicker bones than either of us. Bone cross-sectional thickness increases with use, particularly before the mid-twenties. It suggests significant musculature.

"He was tall, too," Dr. Lieberman said as he held the femur up to my thigh. I'm five foot ten, and the femur was longer than mine. (Dr. Lieberman later sent me a published estimate on Skhul V: five foot ten and 150 pounds. "Yeah, I don't believe it," he said. "I suspect they estimated them wrong.") Even so, five foot ten would have been considered gigantic from the Agricultural Revolution until recently. Early farming populations lost as much as five inches of height compared with early foragers.

"So how long did this guy live?" I asked.

"Well, this guy probably fits the stereotype of dying young," Dr. Lieberman said. "Maybe thirty to forty years old. But life was dangerous back then, and it looks like he was healthy right up to the end. Plenty of ancient hunter-gatherers lived a long time. Contemporary hunter-gatherers regularly live well into their sixties and seventies."

The common misperception is that ancient people would blow out the candles at their thirty-fifth birthday party and then just drop dead. But even chimpanzees and gorillas can live that long in the wild, and there are good reasons to believe we are naturally longer lived than they are. Humans have fewer natural predators than do other primates, as well as a longer childhood before puberty. The age of the oldest documented human (122 years old) far exceeds the oldest documented chimpanzee (66 years old) and gorilla (56 years old), both of which lived in captivity with no risk of predation or starvation. The natural human life span appears to have lengthened in the late Paleolithic when humans were able to fend off external sources of mortality and bear (or support) offspring at older ages. In fact,

it's likely that life expectancy initially *dropped* after the Agricultural Revolution.

The Agricultural Revolution seems like a paradox of history: if human health got worse, then why did people become farmers?

The shift to an agricultural lifestyle wasn't something that anyone consciously planned. Even so, the domestication of animals and plants appears to have taken place independently in multiple locations around the world (the Fertile Crescent, China, India, the Americas, and Africa) at about the same time, 15,000 B.C. to 5,000 B.C. It was a technology whose time had come.

In short, the Agricultural Revolution unlocked a path to more rapid growth—in population, culture, and technology—and the people who took that path left descendants as numerous as the stars in the sky. A fifty-person band of hunter-gatherers—even if they are relatively tall and healthy—will be displaced by a wealthy, technologically advanced city-state with a fast-breeding population, many of whom could be enlisted as soldiers, even if they are short and sickly.

"Shall we go look at some diseased farmers?" asked Dr. Lieberman, and he led me to another part of the archive. He pulled out a couple more boxes of bones and placed them on the counter. He lifted up another skull, holding it for me to have a look. "This is an early Neolithic farmer from Tangier, in modern-day Morocco."

Dr. Lieberman turned it over to show the dental cavity.

"These are some shitty teeth. Once you get farming, you get a lot more cavities. It's classic."

The teeth were ground flat, filled with holes, and many were missing. It looked brutally painful.

"By the way, it's not just starch that causes this. It's also the little bits of stone that get mixed into food when grinding grains. That's why the teeth are all ground flat. And see how shiny they are? They've been polished by the stone bits, like by sandpaper. A lot of starch, plus wear and tear, and a poor diet make for a lot of dental problems."

Dr. Lieberman put the Moroccan farmer away and opened another box.

"We've got lots of diseased farmers. This guy is from the fifteenth century in modern-day Montenegro."

He held up the skull. There were long gaps where teeth were missing. Based on the bones, he wasn't much more than five feet tall.

"This is a squat little person who had a horrible, nasty life," Dr. Lieberman observed. "I know you're focused on diet, but probably the biggest negative impact on the overall health of early farmers came from infectious disease: a result of living in close proximity to other people and domesticated animals for the first time."

What Dr. Lieberman said confirmed my basic take on the story of human health. It wasn't one uninterrupted improvement. *Things got worse before they got better.* Teeth got worse, stature declined, infectious disease went through the roof, and life span declined for the average person.

"Just don't forget that the Agricultural Revolution allowed more people to survive. All of us wouldn't be here without it," Dr. Lieberman said as he packed up the bones.

"Speaking of the Agricultural Revolution, who's hungry for lunch?"

JUST AS with the gorillas I visited at the zoo, understanding a species' wild origins helps make sense of its health needs in more "civilized" environments. So what did the life of a Paleolithic hunter-gatherer look like?

The answer to that question is fraught with academics, whose violent turf wars have earned themselves the name "the fierce people."

A good place to start is by asking anthropologists about contemporary hunter-gatherers. Even though they aren't actually "Paleolithic" peoples, their lifestyles provide a stark contrast with those of most people alive today and are the best living approximation of what life used to be like in the late Paleolithic. Unfortunately, few true hunter-gatherer societies exist anymore. It's down to the Inuit in the Arctic, the Aborigines in Australia, the Hadza in Tanzania, and

various small tribes scattered through remote parts of the Brazilian rain forest, Papua New Guinea, the South Pacific, and Africa. They often live on marginal or remote land and have experienced fairly significant contact with the outside world, often borrowing technology or trading for nontraditional foods.

Historical accounts can provide information about "pre-contact" hunter-gatherer societies that may no longer exist. For example, a Spanish explorer named Álvar Núñez Cabeza de Vaca spent nine years (1528–1537) living with Native American tribes in the modern-day southern United States and northern Mexico. He described the indigenous people as tall and healthy: "[F]rom a distance they look like giants. They are quite handsome, very lean, very strong and light-footed." Unfortunately, few historical observations were made by scientists using hard data, and accounts can be unreliable. Even though Cabeza de Vaca was not a trained anthropologist, his report is in line with more objective sources of evidence, like indigenous skeletal remains in North America.

Archaeologists are a sadistic people who derive pleasure from forcing their grad students to dig a giant hole with a tiny paintbrush. The field is necessarily limited to parts of the body that fossilize (bones, teeth) and objects that don't disintegrate (stone tools, ashes from a fire). Height, bone density, and dental health are straightforward and reliable measurements. Hunter-gatherers from the Upper Paleolithic tended to be taller, had a thicker bone structure, had better teeth, and showed fewer signs of infectious disease than subsequent farmers. But a skeleton doesn't say much about squishy internal organs like the brain, or about important yet intangible qualities like language, culture, or social structure.

Lots of other fields have something to offer. Geneticists are unlocking a wealth of statistical inferences about our ancestors. Primatologists teach us about other closely related species, such as chimpanzees and bonobos. And paleoanthropologists put all the pieces together, grounded in a theoreotical framework based on evolutionary biology.

Combining these disciplines allows us to make an educated guess at what an ancestral human lifestyle might have looked like during the middle to late Paleolithic.

Hunter-gatherer tribes were fairly small—usually fifteen to fifty people. The tribes were nomadic and migrated with prey or sought out available plant life depending on the season. Frequently on the move, they accumulated few material possessions. In warm climates they wore minimal clothing and spent most of their time outdoors.

The primary division of labor was gender based. Men were hunters; women were gatherers. This does not mean that men only hunted animals and women only gathered plants. Women sometimes trapped small game or collected marine foods; men sometimes gathered honey, nuts, or fruit. But men pursued high risk food sources that required greater physical prowess, while women pursued low risk food sources that could be obtained while tending to small children. Food was shared (especially large game), though preference was given to those who played a role in obtaining it. There were rarely any completely specialized professions, other than perhaps a shaman.

The typical hunter-gatherer diet was very diverse. Over the course of a year a diet might have included hundreds of wild plant species and dozens of wild mammals, fish, reptiles, and insects. Almost the entire animal could be eaten or put to use, including bones, organs, and marrow. Roots and tubers were an important food source. The wild predecessors to grains—like wheat, corn, or rice—were negligible until late in the Paleolithic, though some wild grasses were consumed (we ended up domesticating them, after all). Some foods were eaten raw, but a variety of cooking techniques were used. Because of the wide variety of food available and the tribe's migrations, famine was rare to nonexistent.

Hunter-gatherers spent fewer hours procuring food than early farmers did. They walked many miles a day and engaged in more intense physical activities as life called for them: playing, tracking prey, carrying a butchered animal back to camp, digging up roots and tubers, and fighting. Nights were spent around fires, talking, eating,

sleeping, or having sex. Sources of stress were acute (fight, flight) but not chronic (bad boss, credit card debt).

Mothers and other women provided most of the childcare, though mothers were not the exclusive caregiver. Mothers usually breast-fed infants for two to three years. As a result of this longer nursing period, hunter-gatherers had fewer children than did farmers, and spaced them out more. Despite narrowly prescribed gender roles, men and women were on more equitable footing than in subsequent agricultural societies.

Anthropologists have often described hunter-gatherers as optimistic and content. They experienced the full range of emotions, both positive and negative, but negative emotions were rarely chronic. Suicide was rare. They laughed a lot. They had strong personal relationships. They seemed to have a good grasp of what things in life were under their control and what things were not. Their lives appeared to be filled with the meaning that comes from the daily activities required to survive.

There were fewer chronic health problems than in modern industrial society. Obesity and diabetes were nonexistent. Cancer was less common. Dental problems were few, particularly among the young. People had good vision and didn't need eyeglasses or contacts. The human body was well suited to the lives they led.

As pleasant as this may sound, it was no paradise. There was considerable violence. There was widespread infanticide, wife-beating, adultery, rape, and cannibalism. Hunter-gatherers hunted dozens of species to extinction. There were few paths to follow in life. Experience of culture was limited to that of your own tribe and perhaps a few neighboring ones. And, of course, the small-scale nature of hunter-gatherer societies precluded most of the advances of modern civilization.

It wasn't the Garden of Eden.

But compared to the life of an early farmer—which truly was "solitary, poor, nasty, brutish, and short"—a hunter-gatherer existence probably *seemed* like the Garden of Eden.

The biblical description of the Fall of man is one of the oldest human memories. It is the origin tale that lies at the heart of the three great Abrahamic religions: Judaism, Christianity, and Islam. All three sprang from one figure: Abraham, an early herder-farmer. Abraham allegedly lived in the Fertile Crescent—just beyond the only land bridge out of Africa, where the Agricultural Revolution first took off, and where herder-farmers first displaced hunter-gatherers.

Like a dream upon awakening, we struggle to remember life before literacy, life before herding and farming, life before the Fall. But if we step back from the fine details and take in the broad contours, the biblical memory of the Fall has the following arc.

We lived in the Fertile Crescent (Genesis 2:10–14). We lived in harmony with our habitat (Genesis 2:8–25). It did not require much effort to procure food (Genesis 2:8–9). We didn't wear clothes (Genesis 2:25). Then we did something wrong (Genesis 3:6). As punishment, men had to start farming, which was hard (Genesis 3:17–19). We had to eat bread (Genesis 3:19). As punishment, women had to bear more children, childbirth became painful and dangerous, and women fell under the dominion of men (Genesis 3:16). We built the first cities (Genesis 4:17). Our nature now clashed with this new habitat (Genesis 6:5–7, 6:11–12). Agrarian civilizations struggled with famine (Genesis 41), lawlessness (Exodus 20), large-scale warfare (Numbers), and disease (Exodus 7–11). Over time, urban farmers eating plant-based diets displaced hunters and herders eating animal-based diets (Genesis 3:17–19; Genesis 4:2–17; Genesis 25:23–34).

Step even further back, and these early herder-farmers had a memory that goes something like this:

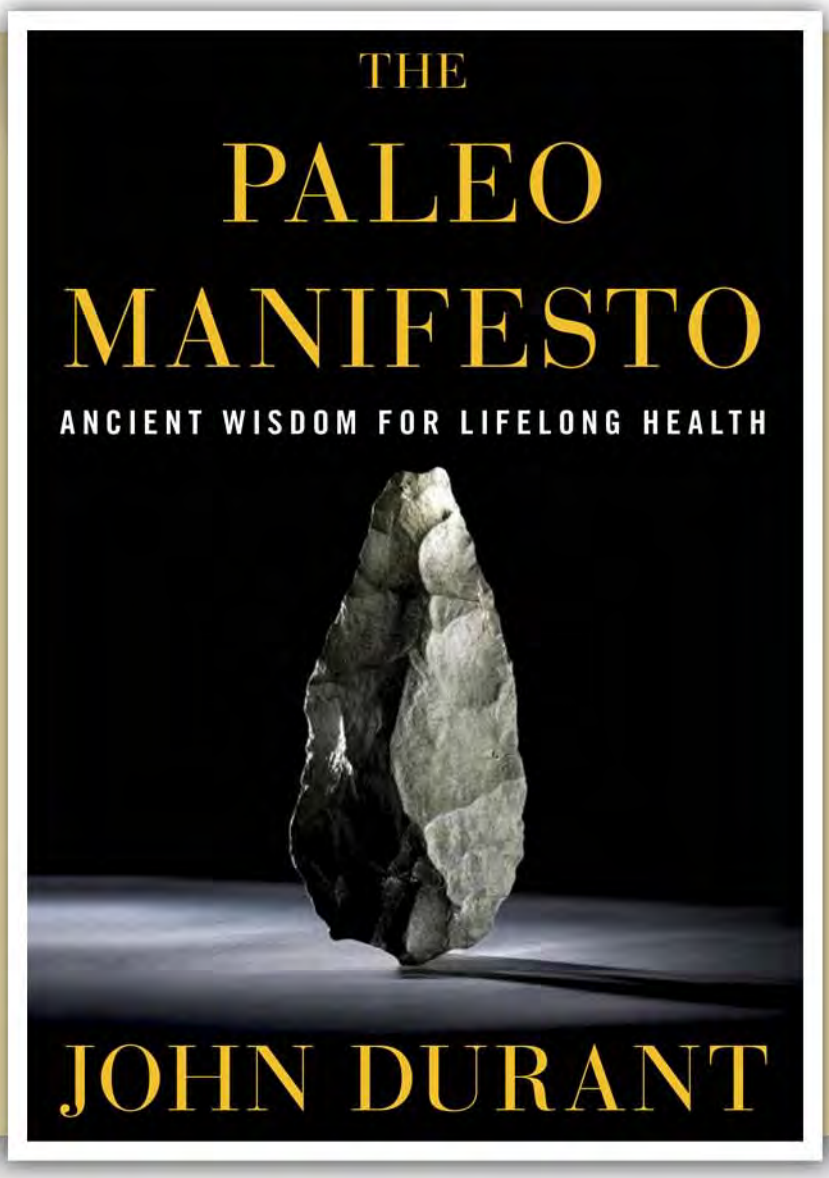
Life was good. We ate something we shouldn't have. Now life is bad.

It would be a decidedly brilliant set of cultural rules that would help a tribe of herder-farmers adapt to life in the early Agricultural Age, and its most important new habitat: the city.

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